

21-SI-Q2

Q (a) NP-hard

(b) distance matrix D

(c) $d(\pi)$ $\pi_1 = \langle 8 \ 4 \ 3 \ 10 \ 7 \ 6 \ 2 \ 5 \ 9 \ 1 \rangle$

$\pi_2 = \langle 5 \ 2 \ 10 \ 3 \ 4 \ 1 \ 9 \ 8 \ 7 \ 6 \rangle$

(d) cycle crossover

(e) $C(\pi)$

(f) $V(\pi)$

Solution

(a) ① NP-hard means problem can be solved but solution can't necessarily be verified within polynomial time

② No known polynomial-time algorithm is guaranteed to find the optimal tour for every instance

(b) distance matrix D , where label 10 correspond Node 0

	0	1	2	3	4	5	6	7	8	9	
0	0.00										
1	7.28	0.00									
2	5.00	12.08	0.00								
3	15.00	14.14	15.81	0.00							
4	24.17	29.41	19.92	21.47	0.00						
5	13.45	20.25	8.49	19.24	13.00	0.00					
6	22.20	29.15	17.20	27.02	12.04	8.94	0.00				
7	20.88	27.29	18.36	34.13	29.73	18.03	19.10	0.00			
8	6.08	11.31	4.24	11.66	18.36	9.49	18.38	22.47	0.00		
9	22.20	24.33	20.25	11.66	13.00	18.38	22.85	36.24	16.49	0.00	

$$d_{i,j} = \sqrt{(x_i - x_j)^2 + (y_i - y_j)^2}$$

$$(c) \textcircled{1} x_1 = \langle 8 \ 4 \ 3 \ 10 \ 7 \ 6 \ 2 \ 5 \ 9 \ 1 \rangle$$

$$\begin{aligned} d(x_1) &= 18.36 + 21.47 + 15 + 20.88 + 19.1 + 17.2 + 8.49 \\ &\quad + 18.38 + 24.33 + 11.31 \\ &= 174.52 \end{aligned}$$

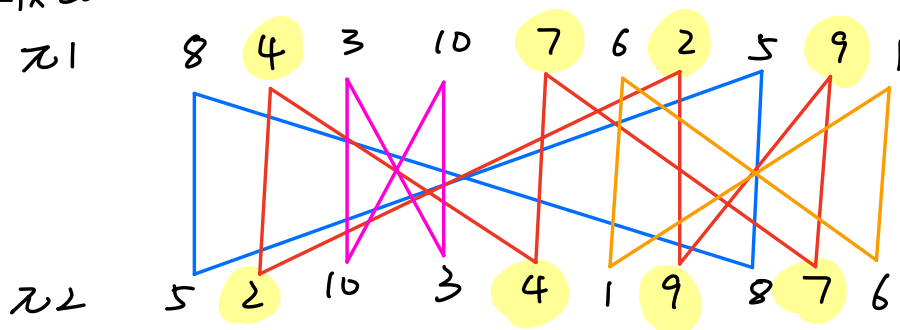
$$\textcircled{2} x_2 = \langle 5 \ 2 \ 10 \ 3 \ 4 \ 1 \ 9 \ 8 \ 7 \ 6 \rangle$$

$$\begin{aligned} d(x_2) &= 8.49 + 5 + 15 + 21.47 + 29.41 + 24.33 + 16.49 \\ &\quad + 22.47 + 19.10 + 8.94 \\ &= 170.7 \end{aligned}$$

cd) cycle cross over

① Cycle detection between the parents gives four cycles: the longest one involves positions: 2, 5, 7, 9

② fixed the 4 7 2 9 position



③ $\pi_1 = \langle 5 \ 4 \ 10 \ 3 \ 7 \ 1 \ 2 \ 8 \ 9 \ 6 \rangle$

$$\pi_2^* = \langle 8 \ 2 \ 3 \ 10 \ 4 \ 6 \ 9 \ 5 \ 7 \ 1 \rangle$$

$$\textcircled{3} \quad d(x_1)^* = 13 + 24.17 + 15 + 34.13 + 27.29 + 12.08 + 4.24 + 16.49 + 22.85 + 8.94 = 178.19$$

$$d(x_2)^* = 4.24 + 15.81 + 15 + 24.17 + 12.04 + 22.38 + 18.85 + 18.03 + 27.29 + 11.31 = 169.12$$

$$f(x_1) = k - 120.5 \quad f(x_2) = k - 170.7$$

$$f^*(x_1) = K - 178.19 \quad f^*(x_2) = K - 169.12$$

$$\textcircled{3} f(x_1) > f^*(x_2) > f(x_2) > f^*(x_1)$$

Child 1 is worse than both parents

Child 2 is better than π_2 , and worse than π_1 .

(e) ① we define $C(\pi)$

$$C(\pi) = \sum_{i=1}^{n-1} a_{\alpha_i, \alpha_{i+1}} + a_{\alpha_n, \alpha_1}$$

② $\pi_1 = \langle 8 4 3 \ 10 \ 7 \ 6 \ 2 \ 5 \ 9 \ 1 \rangle$

$$C(\pi_1) = 6$$

	1	2	3	4	5	6	7	8	9	10
1	0	1	0	1	0	1	0	1	0	1
2	1	0	1	0	1	0	1	0	1	0
3	0	1	0	1	0	1	0	1	0	1
4	1	0	1	0	1	0	1	0	1	0
5	0	1	0	1	0	1	0	1	0	1
6	1	0	1	0	1	0	1	0	1	0
7	0	1	0	1	0	1	0	1	0	1
8	1	0	1	0	1	0	1	0	1	0
9	0	1	0	1	0	1	0	1	0	1
10	1	0	1	0	1	0	1	0	1	0

③ $\pi_2 = \langle 5 \ 2 \ 10 \ 3 \ 4 \ 1 \ 9 \ 8 \ 7 \ 6 \rangle$

$$C(\pi_2) = 8$$

	1	2	3	4	5	6	7	8	9	10
1	0	1	0	1	0	1	0	1	0	1
2	1	0	1	0	1	0	1	0	1	0
3	0	1	0	1	0	1	0	1	0	1
4	1	0	1	0	1	0	1	0	1	0
5	0	1	0	1	0	1	0	1	0	1
6	1	0	1	0	1	0	1	0	1	0
7	0	1	0	1	0	1	0	1	0	1
8	1	0	1	0	1	0	1	0	1	0
9	0	1	0	1	0	1	0	1	0	1
10	1	0	1	0	1	0	1	0	1	0

④ $\pi_1^* = \langle 5 \ 4 \ 10 \ 3 \ 7 \ 1 \ 2 \ 8 \ 9 \ 6 \rangle$

$$C(\pi_1^*) = 6$$

	1	2	3	4	5	6	7	8	9	10
1	0	1	0	1	0	1	0	1	0	1
2	1	0	1	0	1	0	1	0	1	0
3	0	1	0	1	0	1	0	1	0	1
4	1	0	1	0	1	0	1	0	1	0
5	0	1	0	1	0	1	0	1	0	1
6	1	0	1	0	1	0	1	0	1	0
7	0	1	0	1	0	1	0	1	0	1
8	1	0	1	0	1	0	1	0	1	0
9	0	1	0	1	0	1	0	1	0	1
10	1	0	1	0	1	0	1	0	1	0

$$(f) \pi^* = \langle 8 \ 2 \ 3 \ 10 \ 4 \ 6 \ 9 \ 5 \ 7 \ 1 \rangle$$

$$C(\pi^*) = 4$$

	1	2	3	4	5	6	7	8	9	10
1	0	1	0	1	0	1	0	1	0	1
2	1	0	1	0	1	0	1	0	1	0
3	0	1	0	1	0	1	0	1	0	1
4	1	0	1	0	1	0	1	0	1	0
5	0	1	0	1	0	1	0	1	0	1
6	1	0	1	0	1	0	1	0	1	0
7	0	1	0	1	0	1	0	1	0	1
8	1	0	1	0	1	0	1	0	1	0
9	0	1	0	1	0	1	0	1	0	1
10	1	0	1	0	1	0	1	0	1	0

$$(f) \text{ Let } d(\pi) = n - C(\pi)$$

and choose a penalty coefficient

$\lambda > \max d_{ij}$, so that any infeasible tour scores lower than the worst feasible one

$$V(\pi) = K - d(\pi) - \lambda d(\pi)$$